Digital Well-Being for Older Adults: The Impact of Family Neglect on Smartphone Dependence

Abstract: The excessive reliance and addiction of older adults to digital technology are increasingly evolving into a global social and public health challenge. Grounded in social support theory, this study investigates the negative family factors that contribute to compulsive smartphone use among older adults. Using a nationally representative quota sampling approach, we collected survey data from 1,497 older adults aged 60 and above in mainland China. The data were analyzed through partial least squares structural equation modeling (PLS-SEM). The findings reveal that child neglect significantly fosters compulsive smartphone use among older adults, whereas partner neglect has no direct effect. Moreover, perceived family support and loneliness mediate the relationship between child neglect and compulsive smartphone use, with a sequential mediation effect observed between them. In contrast, partner neglect influences compulsive smartphone use only through increased loneliness. This study highlights the differential impacts of child neglect and partner neglect on older adults' psychological well-being and digital consumption habits, providing theoretical and empirical support for developing targeted intervention strategies.

Keywords: compulsive smartphone use, older adults, neglect, loneliness, perceived family support

Introduction

With the development of age-friendly media technologies, an increasing number of older adults are integrating into digital life and enjoying the convenience brought by technological empowerment. According to the Pew Research Center, as of 2023, 75% of Americans aged 65 and above use the internet, a significant increase over the past decade. During this period, the gap in social media use between adults under 30 and those aged 65 and above in the U.S. narrowed from 71 percentage points to 39 percentage points (Faverio, 2022). Similarly, a survey by the UK Office for National Statistics (ONS, 2021) found that by 2020, the proportion of older adults aged 75 and above who frequently use the internet had risen from 29% in 2013 to 54%. In China, data from the China Internet Network Information Center (CNNIC, 2024) indicates that as of August 2024, individuals aged 50–59 and those aged 60 and above accounted for 15.2% and 20.8% of new Internet users, respectively. These statistics reflect a global trend of increasing internet adoption and usage duration among older adults.

Simultaneously, some older adults, like younger users, exhibit compulsive smartphone use behaviors such as smartphone addiction and nomophobia (fear of being without a mobile phone). Data from China indicates that over 10% of middle-aged and older adults (aged 50 and above) spend more than six hours online daily, with some seniors spending an average of over ten hours per day on certain news applications—far exceeding usage levels observed in other age groups. The excessive use of smartphones among older adults has become a pressing global social and public health issue, drawing the attention of scholars from various countries (Busch & McCarthy, 2021; Jia, Liu, & Yang, 2022; Özbek & Karaş, 2022; Liu, Jia, Yang, & Chen, 2023). Therefore, exploring the motivations and underlying causes of compulsive smartphone use among older adults is crucial for enhancing their digital well-being and promoting a healthier lifestyle.

In recent years, research interest in compulsive smartphone use among older adults has grown (Busch & McCarthy, 2021; Allcott, Gentzkow, & Song, 2022; Özbek & Karaş, 2022), with studies examining contributing factors at the individual, family, and societal levels. Among family-related factors, previous research has primarily focused on positive influences such as family support (Jia et al., 2022) and family interaction (Yang, Liu, & Jia, 2022), while relatively little attention has been given to negative family dynamics. Existing studies on adverse family influences have largely examined intergenerational conflicts (Liu, Jia, Yang, & Chen, 2023) and their impact on problematic smartphone use among older adults, yet the potential harm caused by more implicit negative factors such as neglect remains underexplored. Neglect, as a subtle and often unnoticed form of familial disruption, is particularly prevalent in the daily lives of older adults (Lin & Giles, 2013). Therefore, this study focuses on the impact of neglect—both by children and partners—on compulsive smartphone use among older adults.

Existing research suggests that older adults place increasing importance on their relationships with their children in the context of family life (Yan, 2016), while also valuing emotional companionship and deep emotional resonance with their partners (Damianakis, Coyle, & Stergiou, 2020). Accordingly, it is necessary to explore both child neglect and partner neglect as distinct but interconnected dimensions. Based on this premise, this study examines how these two forms of neglect influence compulsive smartphone use among older adults and the underlying mechanisms driving this relationship.

Drawing on social support theory (SST), this study investigates how child neglect and partners affects older adults' psychological states—specifically, perceived family support and loneliness—and how these psychological states, in turn, contribute to compulsive smartphone use. By doing so, this study enriches the academic discourse on compulsive smartphone use among older adults, child and partner neglect, and social support theory. Additionally, it offers valuable insights for policymakers, community workers, and the children of older adults, providing targeted recommendations to address smartphone dependency among older adults and promote their healthy engagement with digital technology.

Literature Review and Hypotheses

Compulsive Smartphone Use

Over the past decade, with the advancement of mobile internet technology and the widespread adoption of smartphones, the issue of smartphone addiction among older adults has increasingly attracted academic attention. Research has shown that compulsive smartphone use among the elderly can have negative impacts on their physical and mental health. Therefore, identifying effective intervention measures to promote the digital well-being of older adults has become an important research direction.

Compulsive smartphone use refers to an individual's uncontrollable over-reliance on smartphones, often accompanied by unhealthy dependency behaviors (B. Chen et al., 2017) and a tendency to struggle with detaching from their devices (Cho & Lee, 2017). This behavioral trait manifests in excessive smartphone usage, even when such usage has already caused significant negative consequences in life (Davis, 2001), such as impairing social interactions, work, learning, and psychological well-being.

As self-regulation capabilities decline in later life, older adults are more vulnerable to compulsive smartphone use and addiction (Busch, Hausvik, Ropstad, & Pettersen, 2021). Conducting in-depth research on the behaviors and underlying mechanisms of compulsive smartphone use among older adults can help in understanding their social participation and psychological needs. This not only provides support for developing policies targeted at the elderly but also optimizes the adaptability of healthcare services, fosters harmonious family relationships, and enhances their sense of social inclusion and well-being.

Social Support Theory

Social Support Theory (SST) is widely used in social sciences to describe the interaction between individuals and their social networks, as well as its impact on psychological and physiological well-being (Underwood, 2000; Dean, Kolody, & Wood, 1990). Social support is defined as assistance provided through social relationships and interpersonal interactions, encompassing four types: emotional support, informational support, instrumental support, and appraisal support. These forms of support help alleviate stress, enhance coping abilities, and promote overall well-being (Cobb, 1976). Among them, emotional support is considered one of the most critical forms of social support, defined as the care, empathy, and psychological comfort provided by others, which is essential for coping with stress and emotional imbalances (House, 1983).

SST has been widely applied in analyzing issues related to elderly social participation, unhealthy behaviors, and mental health (Pan, Feng, & Li, 2019). However, different types of online support (e.g., informational support and emotional support) often influence behavior through different mechanisms (Li, Liao, Kim, Feng, & Pan, 2022). In studies on compulsive smartphone use among older adults, early research focused primarily on instrumental and informational support, exploring how older adults obtain such support through smartphones and pointing out that their excessive reliance on smartphones for information acquisition may lead to compulsive use (Salehan & Negahban, 2013).

In recent years, researchers have increasingly recognized the crucial role of emotional support in elderly internet use (Pan, Feng, Wingate, & Li, 2020). For example, studies have shown that social support can mitigate internet addiction among older adults, with hope and loneliness serving as mediating factors between social support and internet addiction (Jia et al., 2022). Additionally, intergenerational interactions—such as financial support, caregiving, and emotional communication—significantly impact elderly life satisfaction(J. Chen & Jordan, 2018; Schwarz, Oyserman, & Peytcheva, 2010). These studies highlight the key role of social support in improving the quality of life for older adults and reducing unhealthy behaviors.

Children and partners are the most important sources of family support for older adults (Guadalupe & Vicente, 2021). However, systematic research on the specific impact of their support on the psychological state and behavioral patterns of older adults remains insufficient. At the same time, neglect—often regarded as the negative manifestation of social support—is a common trigger for emotional support deficiencies and is widespread in real life. Yet, the differentiated impact of neglect from different family members has not been fully explored.

To address these research gaps, this study, based on Social Support Theory, focuses on emotional support and examines how emotional neglect from partners and children within the support network triggers psychological changes in older adults, and how these changes further influence their unhealthy smartphone use.

Child and Partner Neglect

Neglect is one of the most common forms of elder abuse and is recognized as the most prevalent and far-reaching among the four major types of abuse (WHO, 2008). The World Health Organization (WHO) defines neglect as the "refusal or failure to fulfill caregiving obligations within a relationship of trust" (Wolf, Bennett, & Daichman, 2003). Among its various forms, emotional neglect is particularly significant, referring to the prolonged disregard, belittling, or lack of recognition of an individual's emotional needs (Ludwig & Rostain, 2009). Emotional neglect is especially common in contemporary Chinese families, where family members may fail to meet the basic emotional needs of older adults or exhibit slow or insufficient emotional responses.

Existing research on neglect has largely focused on adolescent populations (Hildyard & Wolfe, 2002; Stoltenborgh, Bakermans-Kranenburg, & Van Ijzendoorn, 2013). However, with the transformation of family structures and the increasing prevalence of elderly individuals living alone, emotional neglect among older adults demands urgent attention (Stein, 1991).

Current studies predominantly examine self-neglect among older adults (Dong et al., 2009; Iris, Ridings, & Conrad, 2010), while research on neglect from family members remains insufficient. Older adults may experience two types of neglect within the family: child neglect and partner neglect. Child neglect refers to the failure of children to provide the necessary emotional support and care for their elderly parents. It manifests in a lack of communication, ignoring the emotional needs of the elderly, or failing to act when support is needed, leaving them feeling isolated and neglected (Koga, Tsuji, Hanazato, Nakagomi, & Tabuchi, 2024). Partner neglect refers to the failure of an elderly individual's partner to provide emotional support and companionship within the relationship. This may involve disregarding the partner's emotional needs, ignoring communication, or failing to respond during significant emotional moments.

Both types of neglect can have severe negative impacts on the mental health of older adults, potentially leading to feelings of loneliness, depression, and diminished self-esteem (Ross, 2023).

Child Neglect, Partner Neglect, and Compulsive Smartphone Use in Older Adults

Compulsive smartphone use (CSU) refers to excessive smartphone use that interferes with daily activities. This study hypothesizes that older adults may develop compulsive smartphone use due to neglect by their children and partners, using it as a coping mechanism to fulfill basic psychological needs and seek social network support.

According to social support theory, older adults require emotional support from family to maintain their psychological well-being and engage in positive behaviors (Özbek & Karaş, 2022). Child neglect is often characterized by reduced communication or lack of interaction with older adults, which can diminish their understanding of family dynamics and, in turn, reduce their confidence in socializing with peers (Charles & Carstensen, 2010). This lack of information may cause feelings of alienation, compelling older adults to seek information or alternative social support through smartphones. Furthermore, child neglect may lead to insecurity regarding their role within the family, prompting frequent smartphone use as a means of social compensation.

In contrast, partner neglect is more closely related to emotional deprivation. Partner neglect can result in deeper emotional distress, such as loneliness, shame, or rejection (Terrizzi Jr & Shook, 2020). These emotions may lead older adults to withdraw from face-to-face interactions and rely on smartphones as a tool for emotional release and avoidance. The instant connectivity provided by smartphones enables them to establish alternative social networks via social media, online gaming, and video calls, thereby alleviating emotional isolation (Cui, Zou, Ji, & Zhang, 2024; Liang, Xiong, & Xie, 2022; Ren & Klausen, 2024). Thus, when neglected, older adults tend to increase their smartphone usage as a coping mechanism to compensate for missing social connections and fulfill psychological needs (Brailovskaia, Stirnberg, Rozgonjuk, Margraf, & Elhai, 2021; Sun, 2023; Holte, Aukerman, Padgett, & Kenna, 2024). However, frequent smartphone use and dependency can eventually develop into compulsive smartphone use.

Based on this analysis, we propose the following hypotheses:

- **H1**: Child neglect positively predicts compulsive smartphone use in older adults.

- **H2**: Partner neglect positively predicts compulsive smartphone use in older adults.

Although prior research has shown that both child neglect (Silverstein, Chen, & Heller, 1996) and partner neglect (Ng, Huo, Han, Birditt, & Fingerman, 2022) have profound effects on the psychological well-being and behavior of older adults, it remains unclear whether the two types of neglect differ significantly in their influence on compulsive smartphone use. Child neglect and partner neglect may impact compulsive smartphone use through different emotional and social support mechanisms.

Firstly, child neglect typically involves a lack of information exchange and social support (Liu, Jia, Yang, & Chen, 2023). This may lead to a reduced sense of social connectedness, increasing older adults' reliance on digital social interactions. Since smartphones provide alternative means of social engagement (e.g., social media and instant messaging), older adults who experience neglect by their children may be more inclined to use smartphones to enhance their sense of social connectedness and compensate for a lack of real-world interactions.

In contrast, partner neglect primarily involves deep emotional deprivation and a lack of intimacy (Kieslich & Steins, 2022). Partners typically serve as long-term emotional supporters, so when older adults experience coldness or neglect from their partner, they may turn to smartphones for emotional solace—watching videos, browsing social networks, or engaging in virtual interactions to seek emotional comfort and reduce feelings of loneliness.

Based on these differences, we propose that child neglect and partner neglect may have different levels of impact on compulsive smartphone use. Child neglect is more likely to drive smartphone use through the social substitution pathway, while partner neglect may contribute to compulsive smartphone use through the emotional regulation pathway.

Therefore, this study further raises the following question to explore:

- **RQ1**: Do child neglect and partner neglect have different degrees of influence on compulsive smartphone use in older adults?

Mediating Role of Perceived Family Support and Loneliness

Among older adults, family support is considered the primary source of social support, reflecting the care, respect, and recognition received from family members (Turner & Turner, 2013). Such support is crucial for psychological well-being and quality of life. Research suggests that as individuals age, their social networks shrink, leading them to prioritize emotional needs in significant relationships (Carstensen, Isaacowitz, & Charles, 1999). As a result, care and companionship from children and partners become increasingly important. When children and partners fail to meet these needs, older adults may experience a decline in self-worth, weakening their perceived family support (Derkx, Bos, Laceulle, & Machielse, 2020).

When older adults perceive insufficient family support, they may turn to the digital world to fill the emotional void. Smartphones provide a means of escaping reality and seeking emotional compensation (Kardefelt-Winther, 2014), for example, by building virtual social relationships or browsing social media to stay connected to the world. In real life, as face-to-face interactions with family members decrease, older adults lose an essential social support system, further increasing their dependence on the digital world and leading to compulsive smartphone use.

Based on this analysis, we propose the following hypotheses:

- **H3**: Perceived family support mediates the relationship between child neglect and compulsive smartphone use.

- **H4**: Perceived family support mediates the relationship between partner neglect and compulsive smartphone use.

Loneliness is closely linked to emotional, physical, and overall well-being. From a relational perspective, loneliness refers to a lack of sufficient social connections and intimate relationships, making it difficult for individuals to adapt to and integrate into society (Perlman & Peplau, 1981).

Older adults typically rely on children and partners for emotional support and companionship, expecting family members to show concern for their lives, listen to their feelings, and guide them through modern life. Such interactions help build close family relationships. However, when neglected by children or partners, older adults may feel excluded from the family system, leading to diminished self-efficacy and self-esteem, which in turn exacerbates loneliness. Neglect can also result in social isolation, making it harder for older adults to maintain stable connections with those around them. At the same time, emotional deprivation caused by neglect further intensifies feelings of loneliness, making it increasingly difficult for older adults to integrate into society.

In response to loneliness, older adults may turn to digital media to seek social and emotional compensation. The internet, social media, and online games provide an escape from reality and a means of fulfilling social needs. By interacting with others or consuming entertainment content, older adults can find emotional solace and experience a sense of attention and belonging. However, this behavior can gradually lead to excessive smartphone dependence, ultimately resulting in compulsive use.

Based on this analysis, we propose the following hypotheses:

- **H5**: Loneliness mediates the relationship between child neglect and compulsive smartphone use.

- **H6**: Loneliness mediates the relationship between partner neglect and compulsive smartphone use.

According to social support theory, an individual's social support system significantly impacts their physical and mental health (Cohen & Wills, 1985). Increased social support helps alleviate negative psychological conditions, thereby reducing loneliness (Holt-Lunstad, Smith, Baker, Harris, & Stephenson, 2015). Empirical research using longitudinal data has confirmed that a lack of family support contributes to greater loneliness in older adults (Y. Chen, Hicks, & While, 2014; Kang, Park, & Wallace, 2018). Therefore, this study predicts that perceived family support, as a crucial social support system, can effectively mitigate loneliness.

Building on this analysis, increased emotional child neglect and partners lead to lower perceived family support, which in turn heightens feelings of loneliness. As a result, older adults may become further addicted to smartphone use.

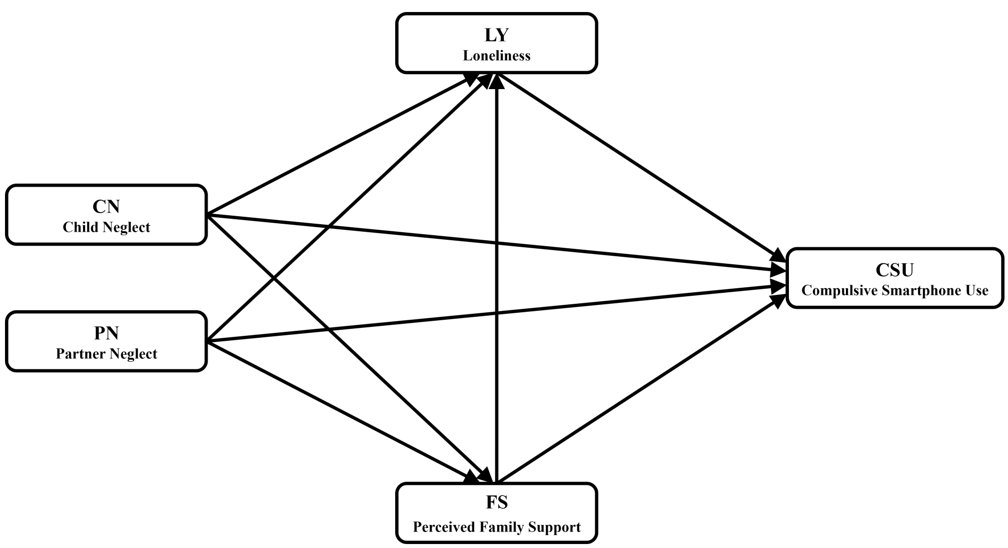

Figure 1
Theoretical model.

Based on this analysis, we propose the following hypotheses:

- **H7**: Perceived family support and loneliness have a chain mediation effect between child neglect and compulsive smartphone use.
- **H8**: Perceived family support and loneliness have a chain mediation effect between partner neglect and compulsive smartphone use.

The theoretical model illustrating these relationships is presented in Figure 1.

Research Methodology

Sample Collection

From February to August 2024, we conducted an online survey using the Tencent Questionnaire Aging Sample Database and obtained 1,497 valid responses. As shown in Table 1, all participants were aged 60 and above and resided in mainland China. The study employed a quota sampling method to ensure representativeness, based on demographic data from the Seventh National Population Census. Participants were stratified by gender, age, educational attainment, and household registration status. However, due to the reliance on online participation, systematic biases may still exist. Additionally, because of historical factors such as war and social upheaval, the educational level of older Chinese adults is generally low, with most respondents having an education level below high school.

For a long time, older adults have been underrepresented in internet use, particularly in traditional online surveys, which typically reach only those with higher levels of internet proficiency. This may lead to selection bias in sample collection. However, the Tencent Questionnaire Aging Sample Database has been dedicated to serving older users for many years, accumulating a rich dataset of 500,000 elderly participants, including both experienced and less-experienced internet users. This broad coverage ensured that the survey reached older adults with varying levels of internet familiarity, enhancing the sample's representativeness and diversity. Furthermore, the sample collection process was approved by our institution's ethics committee.

Table 1

Demographic Information of Participants

Variable	Category	Sample Size	Percentage
Gender	Male	699	46.69%
	Female	798	53.31%
Age	60-64 years old	449	29.99%
	64-60 years old	438	29.26%
	70 years and above	610	40.75%
Education Level	Elementary school or below	872	58.25%
	Middle school	412	27.52%
	High school	148	9.89%
	Associate degree	42	2.81%
	Bachelor's degree or above	23	1.54%
Residence Type	Urban	815	54.56%
	Rural	682	45.44%
Retirement Status	Not retired	424	28.32%
	Retired	1073	71.68%
Annual Income	Below 5,000 yuan	487	32.53%
	5,000-10,000 yuan	224	14.96%
	10,000-20,000 yuan	184	12.29%
	20,000-50,000 yuan	246	16.43%
	50,000-100,000 yuan	201	13.43%
	100,000 yuan and above	155	10.35%

N = 1497

Variable Measurement

All variables in this study—including compulsive smartphone use, child neglect, partner neglect, perceived family support, and loneliness—were measured using a 5-point Likert scale, ranging from 1 (strongly disagree) to 5 (strongly agree).

Compulsive Smartphone Use Scale: The measurement of compulsive smartphone use was adapted from the Elderly Internet Addiction Scale revised by Liu, Jia, Yang, Yan, and Mu (2023). Specifically, this study modified the "Compulsive Internet Use" subscale to assess compulsive smartphone use among older Chinese adults. Sample items include: "I find it difficult to stop using my smartphone" and "I know prolonged smartphone use is harmful, but I cannot control myself." The scale demonstrated good reliability (Cronbach's $\alpha = 0.877$).

Child Neglect Scale: Since no standardized clinical measure exists for evaluating elderly neglect (Strasser & Fulmer, 2007), this study adapted the Emotional Neglect Scale developed by Straus, Kinard, and Williams (1997) to measure child neglect. Sample items include: "My children do not comfort me when I feel sad," "My children rarely express their care for me," and "My children engage in activities with me only for entertainment." The adapted scale exhibited good reliability (Cronbach's $\alpha = 0.850$).

Partner Neglect Scale: The measurement of partner neglect was also adapted from the Emotional Neglect Scale (Straus et al., 1997). Respondents were asked to assess statements such as: "My partner does not comfort me when I feel sad," "My partner rarely praises me," and "My partner participates in activities with me solely for entertainment." The scale demonstrated strong reliability (Cronbach's $\alpha = 0.887$).

Perceived Family Support Scale: Perceived family support was measured using the Perceived Social Support Scale (PSS) developed by Zimet, Dahlem, Zimet, and Farley (1988). Participants evaluated statements such as: "In life, I believe my family will try their best to help me" and "I believe I can discuss problems with my family." The scale demonstrated good reliability (Cronbach's $\alpha = 0.886$).

Loneliness Scale: The measurement of loneliness was based on the ULS-8 Loneliness Scale developed by Hays and DiMatteo (1987) and revised by Zhou, Li, Hu, and Xiao (2012). This is a shortened version of the UCLA Loneliness Scale, with sample items including: "I lack companionship," "I have no one to rely on," and "I feel isolated." The scale demonstrated strong reliability ((Cronbach's $\alpha = 0.876$).

Control Variables: This study controlled demographic variables, including age, gender, education level, household registration status, annual income, and retirement status, as prior research suggests that these factors may influence outcomes.

Analysis Method

This study employed Smart PLS 4.0 for data analysis and hypothesis testing. We chose to use Partial Least Squares Structural Equation Modeling (PLS-SEM) instead of Covariance-Based Structural Equation Modeling (CB-SEM) for the following reasons. First, PLS-SEM demonstrates advantages in handling complex models, especially when the model involves multiple independent variables and mediating variables (Dash & Paul, 2021). Additionally, PLS-SEM is predictive, aiming to maximize the explained variance of the dependent variable (R^2), which makes it more advantageous for predictive analysis and theoretical model construction. Therefore, this study selected PLS-SEM as the primary analysis tool to ensure effective analysis of complex structural models and accurate testing of hypotheses.

Common Method Bias

Due to the simultaneous data collection in cross-sectional studies, there may be common method bias (CMB) that affects the accuracy of the results. To mitigate this risk, several measures were taken during the design and implementation of this study. First, anonymous questionnaires were used, and respondent privacy was ensured to reduce social desirability effects and response

Table 2
Model Evaluation

Variable		Mean	SD	Outer Loadings	Cronbach's Alpha	CR	AVE	VIF
CSU	CSU1	2.411	0.877	0.75	**0.877**	**0.909**	**0.666**	2.34
	CSU2			0.819				2.733
	CSU3			0.843				2.15
	CSU4			0.863				2.34
	CSU5			0.801				1.813
CN	CN1	1.937	0.683	0.8	**0.85**	**0.893**	**0.626**	2.105
	CN2			0.852				2.517
	CN3			0.811				2.47
	CN4			0.755				2.116
	CN5			0.733				1.517
PN	PN1	2.096	0.819	0.854	**0.887**	**0.917**	**0.689**	3.239
	PN2			0.888				3.82
	PN3			0.835				2.778
	PN4			0.792				2.421
	PN5			0.777				1.663
FS	FS1	3.878	0.772	0.844	**0.886**	**0.92**	**0.742**	2.512
	FS2			0.876				3.288
	FS3			0.894				2.986
	FS4			0.831				1.7
LY	LY1	2.136	0.581	0.768	**0.876**	**0.902**	**0.538**	1.864
	LY2			0.788				1.936
	LY3			0.578				1.447
	LY4			0.817				2.211
	LY5			0.625				1.516
	LY6			0.669				1.558
	LY7			0.794				2.071
	LY8			0.791				2.079

CN: Child Neglect; CSU: CSU: Compulsive Smartphone Use; FS: Family Support; LY: Loneliness; PN: Partner Neglect

bias. Second, the questionnaire was designed to avoid ambiguous or double-meaning questions, ensuring clarity and specificity of the items. Additionally, different measurement scales were employed for different latent variables to avoid systematic errors caused by a single source measurement tool. In the data analysis phase, we used SPSS software to conduct Harman's single-factor test to detect potential common method bias (Podsakoff, MacKenzie, Lee, & Podsakoff, 2003). The results showed that the unrotated first factor explained 32.57% of the variance, which is below the critical threshold of 40%, indicating that the impact of common method bias is limited.

Measurement Model Assessment

Table 3
Discriminant validity

Fornell–Larcker criterion					
	CSU	CN	PN	FS	LY
CSU	0.816				
CN	0.263	0.792			
PN	0.226	0.617	0.83		
FS	-0.228	-0.346	-0.247	0.861	
LY	0.278	0.574	0.538	-0.321	0.734
Heterotrait–Monotrait (HTMT) ratio					
	CSU	CN	PN	FS	LY
CSU					
CN	0.289				
PN	0.242	0.713			
FS	0.246	0.38	0.259		
LY	0.292	0.649	0.588	0.342	

CN: Child Neglect; CSU: CSU: Compulsive Smartphone Use; FS: Family Support; LY: Loneliness; PN: Partner Neglect

This study analyzed the reliability, convergent validity, and discriminant validity of five latent variables to assess the model fit. According to Table 2 of the model evaluation, both composite reliability (CR) and Cronbach's alpha (CA) exceeded the critical threshold of 0.70, indicating sufficient construct reliability. Convergent validity was assessed using the average variance extracted (AVE), which was above the critical value of 0.5. Discriminant validity was validated through three methods. First, the Fornell-Larcker test showed that the square root of the

AVE for each variable was greater than its maximum correlation with other variables (see Table 3 for discriminant validity). Second, the factor loadings for each item exceeded 0.5 and loaded onto their respective constructs. Lastly, the Heterotrait-Monotrait ratio (HTMT) analysis indicated that all HTMT values were below the recommended critical threshold of 0.85 (Henseler), confirming good discriminant validity among the constructs. Additionally, we checked for multicollinearity issues by reporting the variance in inflation factors (VIF) values for all constructions in the study. The results showed that all VIF values in Table 2 of the model evaluation were less than 5, indicating that the findings of this study were not significantly affected by multicollinearity.

Structural Model Assessment

The research results as shown in Figure 4 and 5 indicate a significant positive correlation between child neglect and compulsive smartphone use ($b = 0.1, t = 2.48, p < 0.05$), supporting Hypothesis H1. However, the correlation between partner neglect and compulsive smartphone use was not significant ($b = 0.05, t = 1.31, p > 0.1$), meaning that Hypothesis H2 was not supported.

Beyond direct effects, the study also examined perceived family support and loneliness as mediating variables in indirect effects. Using the bootstrapping technique for resampling, the results showed that perceived family support significantly mediated the positive correlation between child neglect and compulsive smartphone use ($b = 0.04, t = 4.47, p < 0.001$), supporting Hypothesis H3. However, perceived family support did not significantly mediate the relationship between partner neglect and compulsive smartphone use ($b = 0.01, t = 1.25, p > 0.1$), thus Hypothesis H4 was not supported. Loneliness was found to mediate the positive relationship between child neglect and compulsive smartphone use ($b = 0.05, t = 4.11, p < 0.001$), supporting Hypothesis H5. Moreover, loneliness also significantly mediated the relationship between partner neglect and compulsive smartphone use ($b = 0.04, t = 3.58, p < 0.001$), supporting Hypothesis H6. Additionally, perceived family support and loneliness acted as sequential mediators between child neglect and compulsive smartphone use ($b = 0.01, t = 2.41, p < 0.05$), supporting Hypothesis H7. However, the sequential mediation effect was not significant for the relationship between partner neglect and compulsive smartphone use ($b = 0.001, t = 0.99, p > 0.1$), meaning Hypothesis H8 was not supported.

Table 4

Hypothesis assessment (direct effect and indirect effect)

Path	Estimate	SE	T-value	P-values	Hypothesis
Direct Effect					
CN->CSU	0.098	0.040	2.476	0.013	H1
PN->CSU	0.051	0.039	1.314	0.189	H2
Indirect Effect					
CN->FS->CSU	0.042	0.009	4.467	0.000	H6
PN->FS->CSU	0.007	0.006	1.246	0.213	H7
CN->LY->CSU	0.053	0.013	4.107	0.000	H11
PN->LY->CSU	0.044	0.012	3.576	0.000	H12
CN->FS->LY->CSU	0.006	0.003	2.410	0.016	H13
PN->FS->LY->CSU	0.001	0.001	0.990	0.322	H14

****p<0.001. CN: Child Neglect; CSU: CSU: Compulsive Smartphone Use; FS: Family Support; LY: Loneliness; PN: Partner Neglect*

Apart from the R^2 values, this study also reported Stone-Geisser's Q^2 values, which measure the predictive relevance of the structural model (Woodside, 2013). The blindfolding technique produced positive Q^2 values for endogenous variables, indicating that the model had good predictive relevance. Furthermore, the Standardized Root Mean Square Residual (SRMR) value was 0.062, which is below the 0.10 threshold, further confirming that the structural model demonstrated a good overall fit.

In addition, to ensure the robustness of the research results, the stability test was also carried

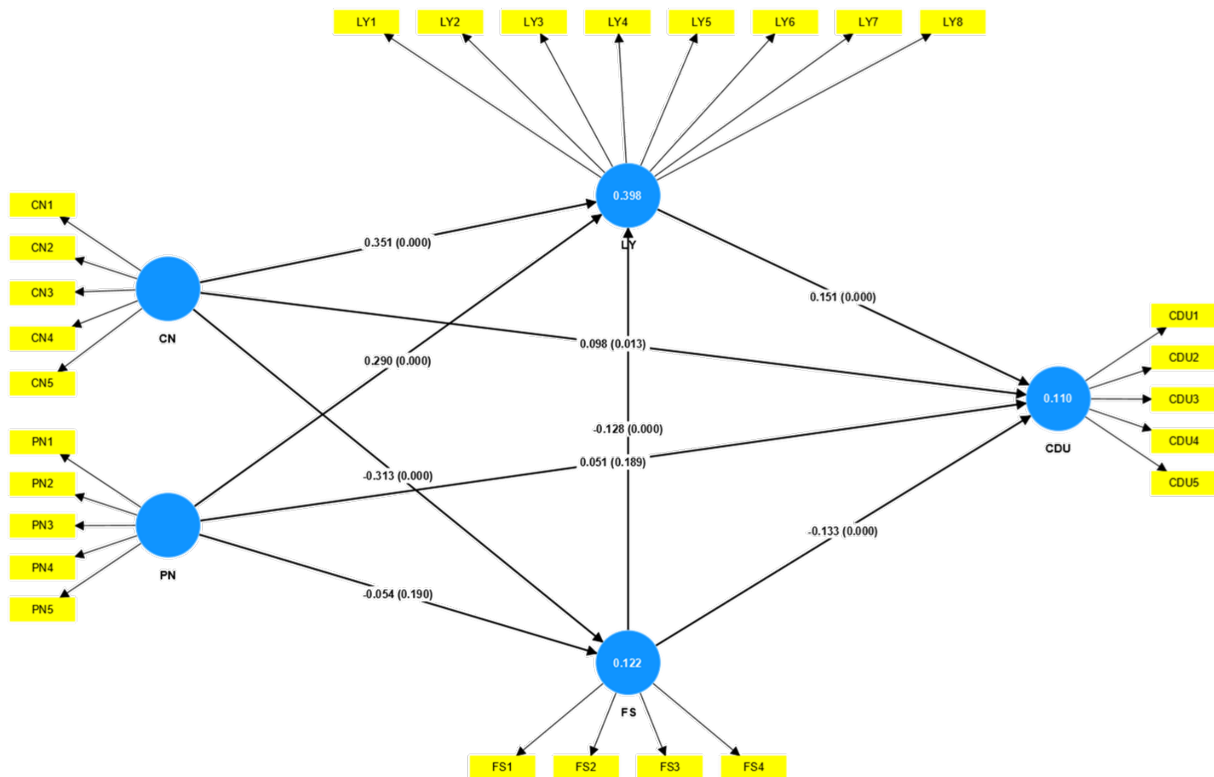

Figure 2
The Result of PLS-SEM.

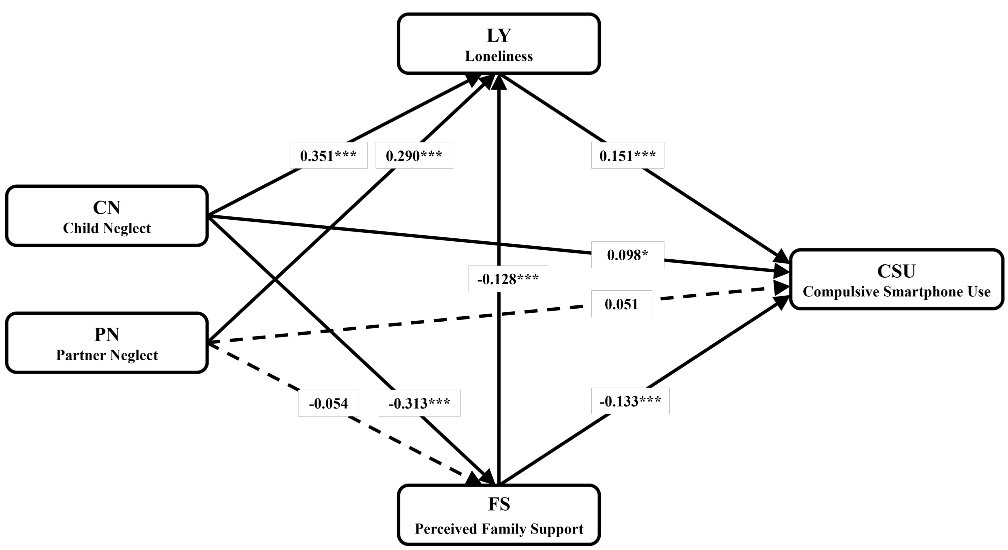

Figure 3
Correlation model map of each variable.

out. Specifically, we excluded the sample of older adults who were divorced, widowed, and childless to avoid potential interference by these circumstances. After removing the relevant samples, the final sample size was 1208. After testing, the regression results of the main variables are basically consistent with the complete sample analysis, indicating that the research conclusion has high stability and universality. This result further validates the reliability of the model and the robustness of the data, providing more solid support for the research findings.

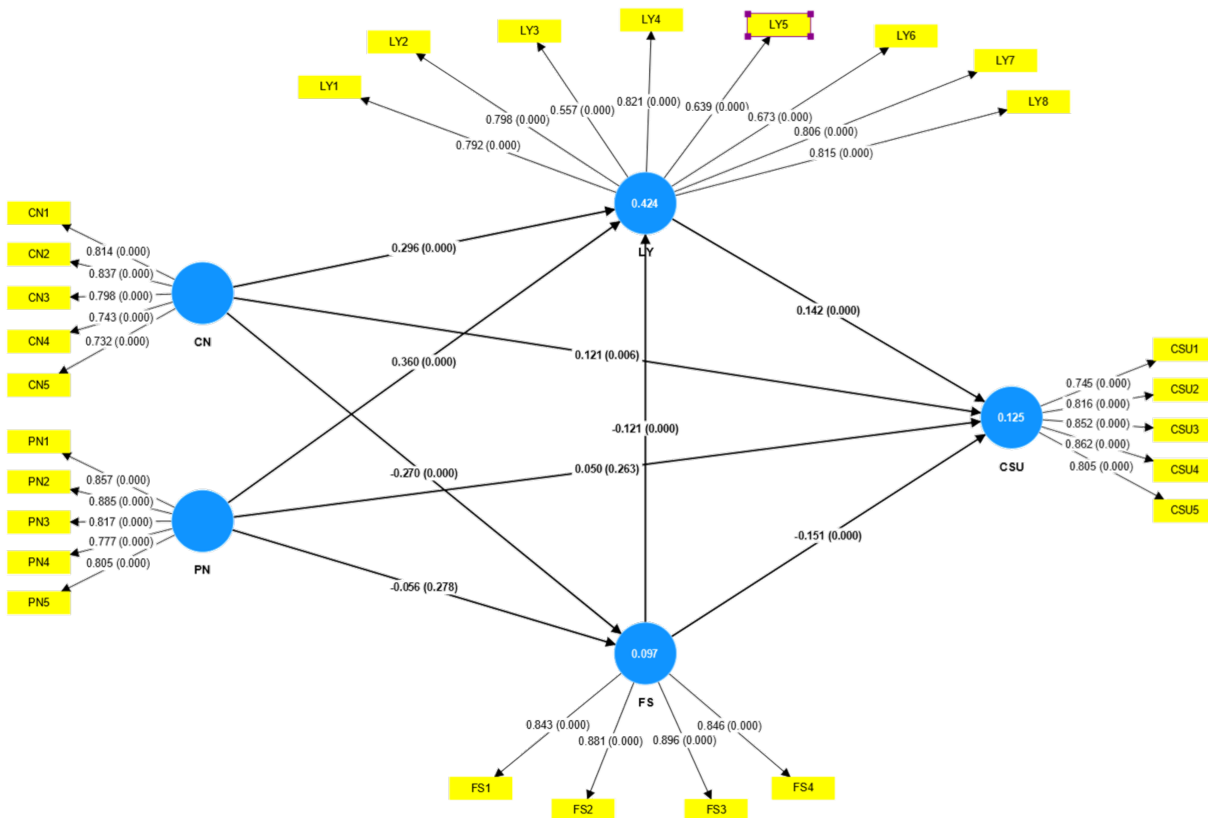

Figure 4
The Result of PLS-SEM in robustness assessment.

Conclusion and Discussion

Based on social support theory, this study explores the complex relationships among child neglect, partner neglect, family support, loneliness, and compulsive smartphone use in older adults. The findings reveal that child neglect not only directly increases the risk of compulsive smartphone use among older adults but also indirectly affects it through inadequate family

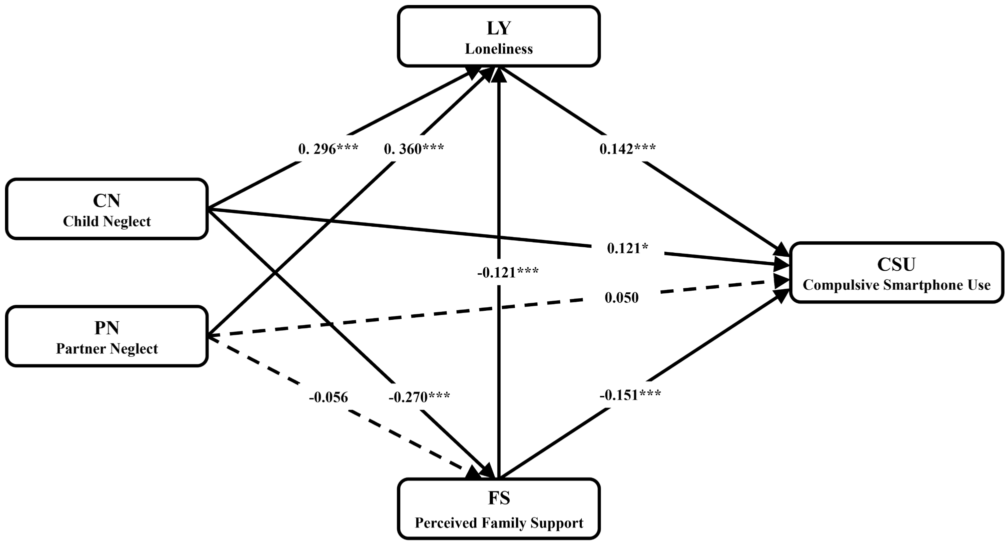

Figure 5
Correlation model map of each variable.

support and loneliness. Furthermore, the study identifies a sequential mediating role of perceived family support and loneliness in the relationship between child neglect and compulsive smartphone use among older adults. However, partner neglect does not have a significant direct impact on compulsive smartphone use or perceived family support in older adults. Instead, partner neglect affects compulsive smartphone use only indirectly by influencing loneliness.

This study addresses research questions and uncovers significant differences in the effects of child neglect and partner neglect on compulsive smartphone use in older adults. Specifically, partner neglect does not have a direct impact on compulsive smartphone use, nor does it significantly affect perceived family support. In contrast, child neglect not only directly affects compulsive smartphone use but also indirectly influences it through family support and loneliness. Additionally, the sequential mediating effect of family support and loneliness in the relationship between child neglect and compulsive smartphone use is particularly prominent.

The insignificant relationship between partner neglect, compulsive smartphone use, and perceived family support in older adults may be attributed to the unique foundations of marriage among older generations and the characteristics of partner relationships. In China, most older adults today were born in the 1950s and 1960s, and their marriages were formed under different social conditions compared to younger generations. These differences may influence their perceptions and responses to partner neglect. The ideology of neo-familism emphasizes prioritizing family interests over individual interests, advocating cooperation among family members, and the maximization of family functionality (Yan, 2021). Specifically, many older adults' marriages were arranged through parental decisions and matchmaking rather than personal choice. Their marriages were often intertwined with social welfare, household registration systems, and housing policies, which might have resulted in a lack of emotional connection based on equality and mutual respect (Coontz & Folbre, 2002). Under such circumstances, Chinese older adults tend to value the stability and practicality of marriage rather than deep emotional exchanges and interactions between partners. Most families operate on a "companionship for daily life" basis, prioritizing the economic and functional aspects of the family as a collective unit (Yan, 2021). Over decades of routine life, older couples may have become accustomed to their interaction patterns, including partner neglect, and learned not to overly rely on emotional support from their partners, instead seeking emotional fulfillment elsewhere. When faced with emotional neglect from their partner, their reactions may be relatively indifferent (Kieslich & Steins, 2022). Consequently, the effect of partner neglect on compulsive smartphone use as a means of seeking social connection and support is not significant. However, it is important to note that partner neglect can still lead to feelings of loneliness in older adults, which, in turn, may drive them to seek emotional solace through excessive smartphone use, potentially leading to compulsive behaviors.

Significant differences exist in how child neglect and partner neglect influence compulsive smartphone use among older adults, which may be linked to differences in expectations for children and partners within Chinese society. Traditional family values in China emphasize that the parent-child relationship is not only a kinship bond but also a psychological and emotional extension (Yan, 2016). In Chinese cultural identity, an individual's sense of self is defined by their social relationships (Fei, Hamilton, & Zheng, 1992). From the moment parents have children, they integrate part of their identity with them. Older adults are often expected to "live for their children", assuming a "sacrificial role" in the family, prioritizing their children's needs, and placing them at the center of their lives, while the role of their partner is often secondary. Under the framework of modern neo-familism, although expectations for children have evolved from absolute obedience ("filial but not submissive") and older adults no longer expect total compliance from younger generations, this shift has not changed the core characteristics of intergenerational relationships (Yan, 2016). Instead, it has led older parents to relinquish potential class advantages in exchange for emotional care and material support from their adult children. Consequently, older adults are likely more sensitive to neglect from their children, experiencing emotional distress that negatively affects their perceived social support. This increases their likelihood of seeking emotional compensation through smartphone and internet use. From a functional and utilitarian perspective, the family serves as the "last safe harbor" for older adults in modern Chinese society. While older adults play a "selfless" role in family reproduction, their selflessness conceals deep emotional expectations for their children. They seek an intergenerational symbiotic relationship, expecting emotional and material support from their children. Therefore, daily care and communication from children are crucial for older adults. Compared to partners, children serve as a more important source of social support. When neglected by their children, the absence of this support has a more profound impact on their psychological well-being and behavioral patterns, leading them to seek emotional compensation through compulsive smartphone use.

Extended Discussion

We find that older adults are not merely passive participants in the digital era; rather, they develop new coping strategies to turn challenges into opportunities for redefining their social roles. As they move through the later stages of life, they not only face weakened family ties and declining social status but also experience “a sense of being outdated” due to technological advancements. However, their reliance on smartphones is not merely a sign of technological adaptation but also a strategy for maintaining social and emotional connections. Traditionally, authority in the family was derived from experience-based knowledge. In the digital era, however, technological proficiency has become the new source of power, shifting the family dynamic so that younger generations guide their parents in adapting to technology rather than the reverse. This shift inverts the power structure, making older adults dependent on their children. However, this dependency is not always negative. Some older adults intentionally avoid learning new technologies to increase their children’s caregiving responsibilities. For example, by claiming to be unable to use smartphones, they request assistance with banking services, online ticket purchases, and other digital tasks, creating more opportunities for parent-child interaction. This strategy not only prolongs intergenerational engagement but also helps older adults maintain influence within the family. Thus, older adults in the digital age are not simply victims; rather, they actively seek ways to adapt and negotiate within the new social structure.

At the same time, despite their efforts to balance family interactions, older adults remain subject to age-related stereotypes, particularly “benevolent discrimination”. Unlike explicit discrimination, benevolent discrimination is more subtle—it appears to offer care and respect but undermines an individual's autonomy. For instance, elder-friendly designs (e.g., mandatory simplified interfaces and large-font modes) seem helpful but often deprive older adults of choice, treating them similarly to minors in need of protection. Such measures disregard their social experience and decision-making capabilities, forcing them into a passive role in society rather than allowing them to integrate fully into the digital world. This hidden regulation of older adults extends beyond technology into social roles. Public "over-care"—such as young people insisting on offering seats even when older adults refuse—imposes societal expectations that "older adults are incapable and must be taken care of". Similarly, age discrimination in the workplace forces many capable older adults into retirement, reinforcing the assumption that they no longer require professional fulfillment, which diminishes their sense of self-worth. Moreover, a double standard in digital consumption further restricts older adults. While young people's engagement with short videos and live streaming is seen as personal freedom, similar behaviors in older adults are labeled as "lack of judgment" or "vulnerability to scams". Under the guise of fraud prevention, society attempts to restrict older adults' internet usage and intervene in their financial decisions, reinforcing the stereotype that they are financially irresponsible and easily deceived.

Ultimately, age-related stereotypes function as a form of social control. While modern society has made significant strides in addressing gender and racial discrimination, ageism remains deeply embedded in social structures. To foster a truly age-inclusive society, it is essential to move beyond a paternalistic approach that disguises control as protection. Instead, older adults should be empowered with genuine autonomy in the digital age, ensuring that they are not merely passive adapters but active participants in shaping their own technological and social experiences.

Theoretical Contributions

This study expands the existing literature on compulsive smartphone use among older adults from multiple perspectives and provides new insights into the impact of family neglect on their compulsive smartphone behaviors.

First, this study reveals that neglect from different family members affects the psychological state and smartphone use behavior of older adults in distinct ways. This finding responds to the argument proposed by Fauth et al. (2024) that family relationships play a crucial role in the mental health of older adults, particularly by further differentiating the impacts of child neglect and partners on compulsive smartphone use. Our findings indicate that child neglect is a significant driving factor behind compulsive smartphone use among older adults. This risk is especially pronounced in the absence of family support. In contrast, partner neglect does not have a significant effect on compulsive smartphone use among older adults. This finding extends the discussion by Roberto and Blieszner (2015) on the stability of marital relationships in later life, suggesting that while partner relationships are important, they do not play a key role in compulsive smartphone use among older adults.

This study proposes a new theoretical framework that clarifies the mediating and sequential mediating roles of family support and loneliness in the relationship between child neglect and compulsive smartphone use among older adults. Our findings support Weiss's theory on the negative effects of loneliness on individual mental health (DiTommaso & Spinner, 1997) and apply it to the context of smartphone use, demonstrating that loneliness is a critical factor driving compulsive smartphone use among older adults. Specifically, insufficient family support exacerbates loneliness in older adults, which in turn triggers compulsive smartphone behaviors.

This finding further supplements the work of Hawkley and Cacioppo (2010) on the interaction between social support and loneliness.

More importantly, this study is the first to reveal that there is no significant correlation between partner neglect and compulsive smartphone use among older adults. This finding aligns with Antonucci, Ajrouch, and Manalel (2017)'s theory on hierarchical relationships in social support networks but provides a more detailed analysis from the perspective of family relationships. It suggests that older adults may have become accustomed to a lack of attention from their partners within their marriage, shifting their emotional dependence more toward their children. Therefore, in interventions targeting compulsive smartphone use among older adults, the emotional support provided by children should be given greater attention, as it plays a crucial role in the digital lives of older adults.

Additionally, this study highlights the cultural specificity of smartphone use among older adults in China. Unlike in Western countries, where older adults primarily use smartphones for information and entertainment, as emphasized by Hunsaker and Hargittai (2018), our findings suggest that Chinese older adults are more driven by a desire for family connection and emotional needs. This discovery provides new empirical support for the argument that social and cultural factors shape digital behaviors among older adults, emphasizing the need to consider diverse sociocultural backgrounds when designing intervention measures.

Practical Implications

The findings of this study suggest several key areas for intervention in digital health and family support policies for older adults. First, given the significant impact of child neglect on compulsive smartphone use among older adults, enhancing parent-child interactions and reducing neglect is crucial. Family members, especially children, should engage more actively in the daily lives of older adults, providing care and companionship to reduce their dependence on smartphones. Community and government agencies can organize family communication training and activities to promote intergenerational communication and reduce emotional estrangement. Second, insufficient family support is a key risk factor for compulsive smartphone use among older adults. Policymakers should actively promote intergenerational support systems, offering more emotional and social support to alleviate loneliness in older adults. Strengthening social services can ensure that older adults receive more emotional care, thereby reducing their reliance on smartphones as a means of filling emotional gaps. Additionally, communities should create more offline social opportunities for older adults by organizing collective activities that help reduce loneliness and the appeal of smartphone use. Meanwhile, businesses can develop digital products with enhanced social functions tailored to the needs of older adults, enabling them to maintain better communication with their children and relatives while minimizing compulsive smartphone use.

Research Limitations and Future Prospects

Despite employing an empirical research approach and uncovering critical insights into the potential impact of child neglect and partners on compulsive smartphone use among older adults, this study has several limitations. First, the cross-sectional nature of the data restricts the ability to explore causal relationships in depth. Future research should adopt longitudinal designs to more accurately capture the long-term effects of child neglect and partners on compulsive smartphone use among older adults and examine its dynamic evolution over time. Second, this study relies on self-reported questionnaires, which may introduce subjective bias and affect the reliability of the results. Additionally, as this research is based on the Chinese context, its global applicability is somewhat limited. Future studies should incorporate multiple data measurement methods and collect data from multiple sources to enhance the objectivity and reliability of findings.

Furthermore, this study focuses only on emotional neglect within family relationships as a negative factor, overlooking other potential family-related issues that may influence digital behaviors among older adults (e.g., inappropriate role distribution, and privacy violations). Such issues may be particularly significant in traditional Chinese family structures, and future research should comprehensively explore various negative family dynamics and their impact on digital behavior among older adults. Finally, to better explain the causes of compulsive smartphone use among older adults, future studies should conduct cross-cultural comparisons, particularly between China—where family is a core social unit—and Western societies. The effects of family neglect on digital behaviors may differ significantly across cultural contexts. Exploring these differences will help deepen our understanding of how cultural backgrounds moderate the relationship between family neglect and digital behaviors among older adults and provide valuable insights for designing culturally adaptive intervention strategies.

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